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Studierichting: Informatica
Oefeningenreeks 5D nr. 4.22

$$y_1 = 2 - \sqrt{x} \text{ en } y_2 = \frac{\sqrt{x} + 1}{2\sqrt{x}}$$

1. Snijpunten

$$2 - \sqrt{x} = \frac{\sqrt{x} + 1}{2\sqrt{x}}$$

$$2\sqrt{x}(2 - \sqrt{x}) = \sqrt{x} + 1$$

$$4\sqrt{x} - 2x = \sqrt{x} + 1$$

$$3\sqrt{x} - 2x - 1 = 0$$

$$t = \sqrt{x}$$

$$3t - 2t^2 - 1 = 0 \iff -2t^2 + 3t - 1 = 0$$

$$t = \frac{-3 \pm \sqrt{9 - 4 \cdot (-2) \cdot (-1)}}{-4}$$

$$t = \frac{-3 \pm \sqrt{1}}{-4}$$

$$t = 1 \text{ of } t = \frac{1}{2}$$

$$\sqrt{x} = 1 \Rightarrow x = 1$$

$$\sqrt{x} = \frac{1}{2} \Rightarrow x = \frac{1}{4}$$

2. Bepaal bovenste functie

$$\text{Neem } x = \frac{1}{2}$$

$$y_1 = 2 - \sqrt{\frac{1}{2}} \approx 1,29$$

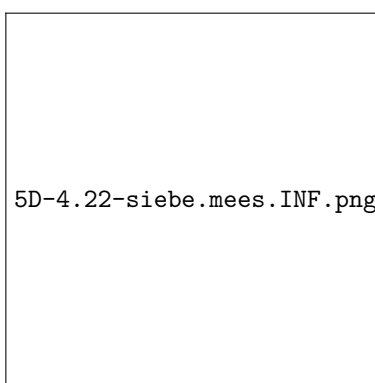
$$y_2 = \frac{\sqrt{\frac{1}{2}} + 1}{2\sqrt{\frac{1}{2}}} \approx 1,21$$

$$\Rightarrow y_1 \text{ boven } y_2$$

$$\begin{aligned}
& \int_{\frac{1}{4}}^1 \left(2 - \sqrt{x} - \frac{\sqrt{x} + 1}{2\sqrt{x}} \right) dx \\
&= \int_{\frac{1}{4}}^1 \left(2 - \sqrt{x} - \frac{1}{2} - \frac{1}{2\sqrt{x}} \right) dx \\
&= \int_{\frac{1}{4}}^1 \left(\frac{3}{2} - \sqrt{x} - \frac{1}{2\sqrt{x}} \right) dx \\
&= \left[\frac{3}{2}x - \frac{2}{3}x^{\frac{3}{2}} - \sqrt{x} \right]_{\frac{1}{4}}^1 \\
&= \left(-\frac{3}{2} + \frac{2}{3} + 1 \right) - \left(-\frac{3}{8} + \frac{2}{3} \cdot \frac{1}{8} + \frac{1}{2} \right) \\
&= \left(-\frac{1}{6} \right) - \left(-\frac{5}{24} \right) \\
&= \frac{1}{24}
\end{aligned}$$

$\Rightarrow$ Tekening

x	0
y_1	2 +
y_1'	—
y_1''	+
y_1	$\searrow$)
x	0
y_2	$+\infty$ —
y_2'	—
y_2''	+
y_2	$\searrow$)



References